解 设以 x 轴正半轴为始边、OA 为终边的角为 θ.

由点 A 的坐标为$(1,2)$，可得 $OA=\sqrt{5}$，$\sin\theta=\dfrac{2\sqrt{5}}{5}$，$\cos\theta=\dfrac{\sqrt{5}}{5}$.

设点 A'的坐标为(x,y)，由 $OA'=OA=\sqrt{5}$，得

$$\begin{aligned}x&=OA'\cos\left(\theta+\frac{\pi}{4}\right)=\sqrt{5}\left(\cos\theta\cos\frac{\pi}{4}-\sin\theta\sin\frac{\pi}{4}\right)\\&=\sqrt{5}\left(\frac{\sqrt{5}}{5}\times\frac{\sqrt{2}}{2}-\frac{2\sqrt{5}}{5}\times\frac{\sqrt{2}}{2}\right)=-\frac{\sqrt{2}}{2},\\y&=OA'\sin\left(\theta+\frac{\pi}{4}\right)=\sqrt{5}\left(\sin\theta\cos\frac{\pi}{4}+\cos\theta\sin\frac{\pi}{4}\right)\\&=\sqrt{5}\left(\frac{2\sqrt{5}}{5}\times\frac{\sqrt{2}}{2}+\frac{\sqrt{5}}{5}\times\frac{\sqrt{2}}{2}\right)=\frac{3\sqrt{2}}{2}.\end{aligned}$$

于是，点 A'的坐标为$\left(-\dfrac{\sqrt{2}}{2},\dfrac{3\sqrt{2}}{2}\right)$.

例 10 把下列各式化为 $A\sin(\alpha+\varphi)(A>0)$的形式：

(1) $\dfrac{1}{2}\sin\alpha+\dfrac{\sqrt{3}}{2}\cos\alpha$；

(2) $\sin\alpha-\cos\alpha$；

(3) $a\sin\alpha+b\cos\alpha(ab\neq0)$.

解 (1) $\dfrac{1}{2}\sin\alpha+\dfrac{\sqrt{3}}{2}\cos\alpha=\sin\alpha\cos\dfrac{\pi}{3}+\cos\alpha\sin\dfrac{\pi}{3}=\sin\left(\alpha+\dfrac{\pi}{3}\right)$.

(2) 因为

$$\sin\alpha-\cos\alpha=\sqrt{2}\left(\frac{\sqrt{2}}{2}\sin\alpha-\frac{\sqrt{2}}{2}\cos\alpha\right),$$

所以

$$\sin\alpha-\cos\alpha=\sqrt{2}\left(\sin\alpha\cos\frac{\pi}{4}-\cos\alpha\sin\frac{\pi}{4}\right)=\sqrt{2}\sin\left(\alpha-\frac{\pi}{4}\right).$$

(3) $a\sin\alpha+b\cos\alpha=\sqrt{a^2+b^2}\left(\dfrac{a}{\sqrt{a^2+b^2}}\sin\alpha+\dfrac{b}{\sqrt{a^2+b^2}}\cos\alpha\right)$.

注意到$\left(\dfrac{a}{\sqrt{a^2+b^2}},\dfrac{b}{\sqrt{a^2+b^2}}\right)$为单位圆上的一点，由正弦及余弦的定义，存在唯一的角 $\varphi\in[0,2\pi)$，使得

$$\cos\varphi=\frac{a}{\sqrt{a^2+b^2}},\quad\sin\varphi=\frac{b}{\sqrt{a^2+b^2}},$$

于是有